VNR Vignana Jyothi Institute of Engineering

Electrical and Electronics Engineering | Course Code: EE301

Power Systems Analysis & Grid Stability - Laboratory Manual & Practical

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Syllabus Focus: EE301 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Power System Modeling & Per-Unit Systems

" Syllabus Scope: Single-line diagrams, per-unit impedances, change of base, synchronous machine modeling

Unit 2: Transmission Line Parameters & Characteristics

" Syllabus Scope: Inductance and capacitance of transposed lines, bundled conductors, short/medium/long l

EE301: Power Systems Analysis & Grid Stability - Advanced Topics & Assessment

Academic Track: Electrical and Electronics Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Power Flow Analysis Load Flow

" Bus classification Slack, PV, PQ, bus admittance matrix Ybus, Gauss-Seidel method, Newton-Raphson method pol

Unit 4: Fault Analysis in Power Networks

" Symmetrical 3-phase faults, subtransient/transient reactances, symmetrical components Fortescue, LG, LL, and

Unit 5: Power System Stability & Equal Area Criterion

" Rotor dynamics, swing equation, power-angle curve, steady-state vs transient stability, Equal Area Criterion

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Formation of Bus Admittance Matrix Ybus and Bus Impedance Matrix Zbus for a 5-Bus Power System

Lab 2: Newton-Raphson Load Flow Analysis and Voltage Profile Determination using MATLAB/Python

Lab 3: Symmetrical and Unsymmetrical Fault Analysis on a Transmission Line Network

Lab 4: Transient Stability Simulation and Critical Clearing Time Determination using Swing Equation

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* John J. Grainger, William D. Stevenson - Power System Analysis McGraw-Hill
* Hadi Saadat - Power System Analysis McGraw-Hill 3rd Ed

Authorized NoteVault Educational Document " Verified for Registered University Students